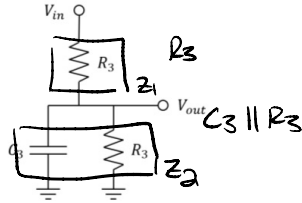


A5 Rivas hux460

Pre-Lab Questions

1.) For the following (LPF) circuit,



Find an expression for the cutoff frequency f_c .

Given that $C_3 = 10\mu F$ what are the different cutoff frequencies for values of $R_3 = [100\Omega, 1k\Omega, 10k\Omega, 100k\Omega]$?

$$f_c = \frac{1}{2\pi(\frac{1}{2}R_3)(C_3)} = \frac{1}{\pi(R_3)(C_3)}$$

100 Ω

$$\Rightarrow \frac{1}{\pi(100\Omega)(10\mu F)} = \underline{318.3 \text{ Hz}}$$

1k Ω

$$\Rightarrow \frac{1}{\pi(1k\Omega)(10\mu F)} = \underline{31.8 \text{ Hz}}$$

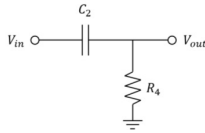
10k Ω

$$\Rightarrow \frac{1}{\pi(10k\Omega)(10\mu F)} = \underline{3.18 \text{ Hz}}$$

100k Ω

$$\Rightarrow \frac{1}{\pi(100k\Omega)(10\mu F)} = \underline{0.31 \text{ Hz}}$$

2.) For the following (HPF) circuit,



Find an expression for the cutoff frequency f_c .

Rearrange the expression used to calculate the cutoff frequency to instead form an expression that calculates the value of the resistor R_4 .

Given that the desired cutoff frequency is $f_c = 1 \text{ kHz}$, what are the different values of R_4 when the C_2 component takes the values $C_2 = [10 \text{ pF}, 0.01 \text{ } \mu\text{F}, 10 \text{ } \mu\text{F}]$?

Instead of using a $16 \text{ k}\Omega$ resistor, assume we choose the values to be set to $C_2 = 0.01 \text{ } \mu\text{F}$ and $R_4 = 15 \text{ k}\Omega$, then what is the new value of f_c ?

$$f_c = \frac{1}{2\pi(R_4)(C_2)} \Rightarrow R_4 = \frac{1}{2\pi(f_c)(C_2)}$$

$$R_4 @ 10 \text{ pF } 1 \text{ kHz}$$

$$R_4 = \frac{1}{2\pi(1 \text{ kHz})(10 \text{ pF})} = \underline{15.9 \text{ M}\Omega}$$

$$R_4 @ 0.01 \text{ } \mu\text{F } 1 \text{ kHz}$$

$$R_4 = \frac{1}{2\pi(1 \text{ kHz})(0.01 \text{ } \mu\text{F})} = \underline{15.9 \text{ k}\Omega}$$

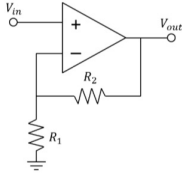
$$R_4 @ 10 \text{ } \mu\text{F } 1 \text{ kHz}$$

$$R_4 = \frac{1}{2\pi(1 \text{ kHz})(10 \text{ } \mu\text{F})} = \underline{15.9 \text{ }\Omega}$$

$$\text{new } f_c \text{ w/ } 16 \text{ k}\Omega \text{ \& } 0.01 \text{ } \mu\text{F}$$

$$f_c = \frac{1}{2\pi(16 \text{ k}\Omega)(0.01 \text{ } \mu\text{F})} = \underline{994.71 \text{ Hz}}$$

3.) Given the following circuit,



What value of resistance is needed for R_2 in order to provide a gain of 101 if $R_1 = 1k\Omega$?

If instead $R_2 = 10k\Omega$ (with $R_1 = 1k\Omega$), then what would the gain of the system be?

For the cascaded systems (i.e., cascaded amplifiers), where A_{sys} has a gain of 10 and B_{sys} has a gain of 10, what is the total gain provided by the cascaded system?

$$\text{Total gain} = A_{sys} \cdot B_{sys}$$

$$= 10 \cdot 10$$

$$= \underline{100}$$

$$\frac{V_{out}}{V_{in}} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1}$$

$$A_v = 1 + \frac{R_2}{R_1} \quad R_1 = 1k\Omega$$

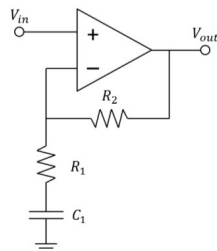
$$\nearrow \quad 101 = 1 + \frac{R_2}{1k\Omega} \rightarrow \underline{R_2 = 1k\Omega \cdot (101 - 1) = 100k\Omega}$$

$$R_2 = 100k\Omega$$

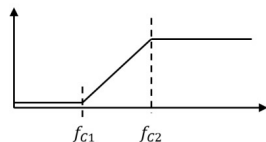
new gain w/ R_2

$$A_v = 1 + \frac{100k\Omega}{1k\Omega} = \boxed{11}$$

4.) Given that the circuit,



has the following approximated frequency response,



Find an expression for the cutoff frequency f_{C2} .

Given that we desire a cutoff frequency of $f_{C2} = 1 \text{ kHz}$, and that $R_1 = 1 \text{ k}\Omega$, what value of C_1 is needed?

Instead, assume we choose the values to be set to $C_1 = 0.1 \mu\text{F}$ and $R_1 = 1 \text{ k}\Omega$, then what is the new value of f_{C2} ?

Find an expression for the cutoff frequency f_{C1} .

What is the cutoff frequency f_{C1} if $C_1 = 0.1 \mu\text{F}$, $R_1 = 1 \text{ k}\Omega$, and $R_2 = 10 \text{ k}\Omega$?

What is the cutoff frequency f_{C1} if $C_1 = 0.1 \mu\text{F}$, $R_1 = 1 \text{ k}\Omega$, and $R_2 = 100 \text{ k}\Omega$?

$$f_{C2} = \frac{1}{2\pi R_1 C_1} \rightarrow C_1 = \frac{1}{2\pi f_{C2} R_1}$$

$$1 \text{ kHz} = 1 \text{ k}\Omega$$

$$C_1 = \frac{1}{2\pi (1 \text{ kHz})(1 \text{ k}\Omega)} = \underline{159.1 \text{ nF}}$$

$$f_{C2} = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi (1 \text{ k}\Omega)(0.1 \mu\text{F})} = 1.59 \text{ kHz}$$

$$f_{C1} = \frac{1}{2\pi (R_1 + R_2) C_1}$$

$$10 \text{ k}\Omega + 1 \text{ k}\Omega = 11 \text{ k}\Omega$$

$$f_{C1} = \frac{1}{2\pi \cdot (11 \text{ k}\Omega)(0.1 \mu\text{F})} = \underline{14.47 \text{ Hz}}$$

$$1 \text{ k}\Omega + 100 \text{ k}\Omega = 101 \text{ k}\Omega$$

$$f_{C2} = \frac{1}{2\pi (101 \text{ k}\Omega)(0.1 \mu\text{F})} = \underline{1.5 \text{ Hz}}$$